

# REDACT: Recoverability Decomposition and Auditable Co-Deletion for the Right to Be Forgotten in LLM Agent Memory

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## Abstract

Ask a paging LLM agent to forget a fact and it may scrub the working-memory block it reasons over while silently leaving the fact in its archival store: the deletion is acknowledged, looks complete, and is not. More generally, deleting a fact from an LLM agent’s memory does not mean the agent has forgotten it. Existing audits check whether a forget command removes a value from retrieval, which is *necessary but not sufficient*, because the value can persist in a derived artifact or be re-derived from facts that were never deleted. We formalize *deletion-completeness* for agent memory, decomposing post-deletion recoverability into three causally distinct channels (residual survival, re-derivation from surviving facts, and a parametric floor recoverable from the base model alone) and give a planner that computes the minimal co-deletion closing the deletable channels, emitting an auditable certificate. Because the parametric floor cannot be deleted, certification has a hard limit: under the strongest modeled adversary, **30 of 81** facts cannot be certified erased (at threshold  $\tau=0.1$ ), even with zero residual survival and no helpful co-deletion. The same stack, run across three memory architectures (a deduplication pipeline, a bi-temporal knowledge graph, and a paging agent), finds residual survival in all three through different by-design mechanisms: the gap is a property of how agent memory derives and retains facts, not of any single system.

## 1 Introduction

A user invokes their right to be forgotten (European Parliament and Council of the European Union 2016) and asks an LLM agent to delete their home address. A modern agent with long-term memory (a paging architecture such as MemGPT/Letta) acknowledges and rewrites the working-memory block it is currently reasoning over. But the address was also written to the agent’s archival vector store in an earlier session; reasoning only over its working context, the agent never touches it, and weeks later an ordinary retrieval surfaces the address again. The deletion was confirmed, looked complete, and was not.

This is not a one-off bug. Even a *faithful* delete of a record need not erase the fact it carried: the fact can survive in a derived artifact the system built from it (a duplicate, a summary, a consolidated note) or be re-derived from facts that remain, such as a salary from a job title plus a public pay band, or

a diagnosis from stored symptoms. The most thorough current audits of agent forgetting (Yang 2026) check whether a forget command removes a value from top- $k$  retrieval. That check is *necessary but not sufficient*: it does not ask *why* a deleted fact returns, nor *what minimal further deletion* would actually close the leak. Those two questions are the subject of this paper.

We cast deletion-completeness as whether a deleted fact remains recoverable, and decompose recoverability into three causally distinct channels: residual survival (a surviving artifact still contains the value), re-derivation (surviving facts entail it), and a parametric floor (the base model recovers it from world knowledge alone). The first two are closable by deleting more; the third is not. A greedy planner computes the minimal additional co-deletion that closes the closable channels, and a certificate records what deletion achieved, at what collateral cost, and when the parametric floor makes erasure impossible to certify. Concretely, we contribute:

- a formal, *adversary-relative* deletion-completeness criterion for agent memory, strictly stronger than dependency-bounded database erasure, decomposing recoverability into residual, re-derivation, and parametric channels (§3);
- a minimal co-deletion planner for the resulting NP-hard problem, reaching completeness at mean collateral  $k = 0.90$  with zero spurious deletions, plus an auditable certificate that separates what deletion achieved from what it cannot (§4, §5.1);
- a *measured* parametric floor and its limit result: under the strongest modeled adversary, 30/81 facts cannot be certified erased even at zero residual survival. Measured as a recovery rate over 8 samples per fact, the count is a Monte-Carlo estimate stable to  $\pm 1$ –2 facts at the  $n=8$ -sample boundary (§5.2);
- evidence of generality: the same probe/planner/certificate stack, run on three memory architectures, exposes residual survival in all three through different by-design mechanisms, including the agent-loop failure above that prior audits bypass (§5.4).

We call the resulting framework REDACT: **R**ecoverability **D**ecomposition and **A**uditable **C**o-**D**eletion.

The breadth across systems is evidence for a stronger, causally-decomposed notion of erasure. This is not a claim to be first to evaluate these systems, since ForgetEval (Yang

2026) already benchmarks all three, but a stronger question asked of them. What is new is asking, and answering, why deleted facts return and what it costs to make them stay gone.

## 2 Related Work

**Auditing forgetting in agent memory.** Agent memory is now routinely benchmarked for long-term *retention*, whether an assistant recalls facts across long histories (Maharana et al. 2024; Wu et al. 2025); its inverse, whether a fact can be reliably *forgotten*, is far less examined. The closest prior work is FORGETEVAL (Yang 2026), the most thorough existing audit of forgetting in deployed agent memory: a careful retrieval-based protocol spanning Mem0, Graphiti, Letta, and roughly a dozen further systems. We build directly on it. Its retrieval-based criterion is the *necessary-but-not-sufficient* gap of §1. Our contribution is therefore the stronger, channel-decomposed criterion, not the system coverage ForgetEval already establishes; correspondingly, our Mem0 figures (recoverability by *any* channel) are complementary to its top- $k$  forget-command rates, not in tension with them. Two of our results extend it directly. ForgetEval deliberately bypasses Letta’s agent loop (addressing archival endpoints to keep the model out of the recall path), which is exactly the loop we exercise and find surface-incomplete (§5.4); and its own per-system observations (Graphiti shedding synthesized fact strings, Mem0’s router favoring link-and-keep) name the mechanisms our decomposition isolates, independently corroborating the convergence result.

**Unlearning methods.** Classical machine unlearning removes a training example by retraining or editing model weights (Bourtole et al. 2021; Cao and Yang 2015; Nguyen et al. 2022). AGENTIC UNLEARNING (Wang et al. 2026) carries this to agents, separating parametric from memory recovery (its “backflow”), but still assumes *weight access* to edit the model; we operate black-box and memory-store-only, so we neither claim the parametric/memory split as novel nor perform model surgery.

**Deletion under dependencies.** Our formal grounding is the database line on erasure when records are inter-derivable: P2E2’s meaningful-erasure criterion (Chakraborty et al. 2025) and inference-aware deletion (Chakraborty et al. 2026). We port P2E2’s *dependency-bounded* criterion to agent memory and deliberately *strengthen* it (its baseline keeps pre-existing operands present, which is vacuous for the right to be forgotten) while inheriting its NP-hardness. This bridge, from relational deletion theory to LLM agent memory, is one the agent-memory works above do not make, and it is where the next section begins.

## 3 Problem Formulation

**Setup.** Let  $f$  be a target fact with value  $v$ , and let  $\Sigma(v)$  be its set of ground-truth surface forms, the ways  $v$  can be written. A memory store  $S$  is a set of records, each carrying text; write  $\text{str}(r)$  for the set of surface strings that appear in record  $r$ . Deleting from  $S$  yields a surviving store  $S' \subseteq S$ . An adversary  $\mathcal{A}$  has full read access to  $S'$ , black-box access to a base model  $M$ , and general world knowledge  $W$ ; a query

$q_f$  asks for  $v$ . We write  $\text{rec}(a) = 1$  when an answer  $a$  reveals a surface form in  $\Sigma(v)$ , as scored by the validated recovery judge (§5.6). Certification is relative to a tolerance  $\tau \in (0, 1)$ ; we use  $\tau=0.10$  (§5). We ask whether a deleted fact is still recoverable, and we split recoverability into three channels that fail for different reasons. Two of them depend on the adversary’s reasoning model, so recoverability is *adversary-relative* (Def. 4).

**Definition 1 (Residual survival).**  $\sigma_{\text{res}}(f, S') = \mathbb{1}[\exists r \in S' : \Sigma(v) \cap \text{str}(r) \neq \emptyset]$ : some surviving record still physically contains a surface form of  $v$ . This channel does not depend on the adversary.

**Definition 2 (Re-derivation).**  $\sigma_{\text{red}}^{\mathcal{A}}(f, S') = \text{rec}(\mathcal{A}(q_f \mid S', M, W))$ : the adversary reconstructs  $v$  by reasoning over the surviving store together with world knowledge, even when no record states  $v$  outright.

**Definition 3 (Parametric floor).**  $\rho^{\mathcal{A}}(f) = \text{rec}(\mathcal{A}(q_f \mid \emptyset, M, C))$ , where  $C \subseteq W$  is the subject’s world-knowable context and the store is *empty*. Because  $\rho^{\mathcal{A}}$  never reads  $S$ , it is system-independent: it measures what the model recovers had  $f$  never been stored at all. We estimate it as a recovery rate over stochastic samples.

**The three channels, in words.** A deleted home address that survives verbatim in a leftover summary note is residual survival ( $\sigma_{\text{res}}$ ). A deleted salary that an adversary rebuilds from a surviving job title plus a public pay band is re-derivation ( $\sigma_{\text{red}}$ ). A deleted “holds a Class-3 licence,” whose implication “at least 18” any model already knows on its own, is the parametric floor ( $\rho$ ). Deleting more shrinks the first two; it cannot touch the third. Figure 1 shows how the probe, planner, and certificate act on each channel.

**Definition 4 (Recoverability, adversary-relative).**  $R_{\mathcal{A}}(f, S') = \max(\sigma_{\text{res}}, \sigma_{\text{red}}^{\mathcal{A}}, \rho^{\mathcal{A}})$ . A fact is as recoverable as its most open channel, since a rational adversary uses whichever one is available; residual is the single channel that does not depend on  $\mathcal{A}$ . The true worst case is  $\sup_{\mathcal{A}} R_{\mathcal{A}}$ , which we bracket with four concrete adversaries spanning prior-generation and frontier models:  $\mathcal{A}_{\text{mini}}$  (gpt-4o-mini),  $\mathcal{A}_{4o}$  (gpt-4o),  $\mathcal{A}_{S5}$  (Claude Sonnet 5), and  $\mathcal{A}_{G5.5}$  (GPT-5.5). A diverse panel matters because adversary strength is not one-dimensional: the frontier models re-derive benign facts more readily, yet they refuse sensitive attribute inferences, so which adversary is worst varies by fact (§5.2).

**Definition 5 (Floor-reaching and erasure-certification).** A deletion is *floor-reaching* for  $\mathcal{A}$  when  $\sigma_{\text{res}} = \sigma_{\text{red}}^{\mathcal{A}} = 0$ : recoverability has been driven down to the parametric floor, and nothing further can be deleted to help. It is *erasure-certified* when it is floor-reaching for every modeled adversary and the floor itself is below tolerance,  $\max_{\mathcal{A}} \rho^{\mathcal{A}} < \tau$  (the sup of Def. 4). These two come apart in exactly one way, and that gap is the limit result of the paper. Floor-reaching is always achievable: co-delete operands until  $\sigma_{\text{red}} = 0$ . Certification, by contrast, fails whenever  $\rho^{\mathcal{A}} \geq \tau$ , because no co-deletion can lower a floor that never depended on the store. A fact can then be as erased as deletion can make it

and still remain world-inferable (R11, Table 2). Reporting the two predicates separately is exact, and it avoids the misleading verdict “complete but recoverable.” This baseline (world knowledge only, every stored operand removed) is deliberately stronger than the dependency-bounded erasure of Chakraborty et al. (2025), which we discuss in §2.

**The optimization problem (Opt-P2E2).** Let  $R_f \subseteq S$  be  $f$ ’s own records. We seek the smallest co-deletion that makes  $f$  floor-reaching:

$$\min_{D \subseteq S \setminus R_f} |D| \quad \text{s.t.} \quad \sigma_{\text{red}}^A(f, S \setminus (R_f \cup D)) = 0,$$

where  $|D| = k$  is the collateral cost; erasure-certification additionally requires  $\rho^A < \tau$  (Def. 5). The co-deleted set  $D$  is made of entailers the user never named, which can read as a data-minimization violation, but it is the opposite: leaving them is not minimization but failed erasure, since  $f$  simply re-derives. The planner minimizes  $k$ , the floor  $\rho$  bounds what no deletion can remove, and because the certificate reports  $k$ , paying it (strong erasure) or not (data minimization) becomes an explicit, audited policy choice rather than a hidden side effect.

**Proposition 1 (NP-hardness).** OPT-P2E2 is NP-hard, by reduction from MINIMUM HITTING SET (Karp 1972; Garey and Johnson 1979). Given a universe  $U$  and sets  $S = \{S_1, \dots, S_m\}$  with  $S_j \subseteq U$ , build an instance with one fact  $x_u$  per element  $u \in U$  and, for each  $S_j$ , an entailment  $\bigwedge_{u \in S_j} x_u \vdash f$ , so each  $S_j$  is a re-derivation path; set  $\sigma_{\text{res}} = 0$  and  $\rho = 0$ . The surviving store re-derives  $f$  exactly when some path  $S_j$  survives intact, so a co-deletion  $D$  achieves  $\sigma_{\text{red}} = 0$  exactly when  $D$  meets every  $S_j$ , that is, when  $\{u : x_u \in D\}$  is a hitting set of  $S$ . The minimum co-deletion therefore equals the minimum hitting set, which is NP-hard. Vertex Cover is the 2-uniform special case, which is exactly our dataset’s pairwise-entailment regime (for instance, F040 re-derived from C001  $\wedge$  C002), so the hardness already holds when every fact has just two operands: it is intrinsic, not an artifact of high-arity rules. The relational hardness of Chakraborty et al. (2025) points the same way. We therefore adopt the greedy, entailment-confidence-ordered planner of §4, which stays exact on the small dependency graphs seen in practice (mean  $k = 0.90$ , §5.1).

## 4 Method

Our pipeline instantiates each object of §3 as a concrete, reusable component (Figure 1) behind a single `MemorySystemAdapter` interface, so the identical probe/planner/certificate stack runs unchanged on all three systems.

**Probe battery.** Three probes measure the three channels of Def. 4. The *exact-match* probe scans every surviving record for a surface form of  $v$  ( $\sigma_{\text{res}}$ ). The *re-derivation* probe gives the adversary the full surviving store as notes (plus today’s date and general world knowledge) and elicits  $v$  in a value-first format ( $\sigma_{\text{red}}^A$ ). The *parametric* probe withholds the store entirely, conditioning only on the subject’s world-knowable

context, and estimates  $\rho^A$  as a recovery rate over  $n$  stochastic samples. The re-derivation *probe* reads the full surviving store (the threat model) and is distinct from the operands-only *injection control* of §5.3, which instead never stores the target value so as to isolate  $\sigma_{\text{red}}$  at  $\sigma_{\text{res}} = 0$ . Recovery is decided by the validated judge (§5.6): a substring match on the ground-truth surface forms, falling back to an LLM adjudication for numeric/semantic approximation. Three auxiliary probes support specific results: a *paraphrase* probe (residual by implication), a retrieval-score *membership-inference* probe (§6), and a *KG-summary* probe for the graph system (§5.4).

**Entailment detector.** Re-derivation paths are found by an LLM-as-judge that scores, for a candidate surviving fact (or set), the confidence that it entails  $v$  given world knowledge, returning a value in  $[0, 1]$  (PARTIAL/NO are capped so a near-miss cannot read as full entailment). The detector both draws the dependency edges and orders co-deletion. Its validation (§5.6) dictates the choice of `gpt-4o` here, since a judge that false-fires on partial operands inflates the collateral.

**Co-deletion planner.** Against a *naive* baseline that deletes only the single surfaced record (no co-deletion; §5.5), we implement two co-deletion heuristics. *Threshold* (the method) probes, purges the target’s own artifacts using the narrow deletion surface form, then co-deletes entailing facts in descending entailment confidence, re-probing after each until  $\sigma_{\text{red}}^A < \tau$ , making  $f$  floor-reaching with a greedily minimal  $D$ . *Depth-first* (an aggressive comparator) co-deletes every entailer above  $\tau$  in one shot, with no re-probe stopping rule. The reported collateral  $k = |D|$  is the measured price of the strong erasure notion (§3); the threshold heuristic minimizes it ( $k = 0.90$  vs. depth-first’s 6.05, §5.1).

**Certificate emitter.** For each target the emitter records the three channel scores, the final recoverability  $R_A = \max(\cdot)$ , the heuristic, the co-deleted set and its size  $k$ , and the threat model. It exposes *two* machine-readable predicates from Def. 5: `floor_reaching` ( $\sigma_{\text{res}} = \sigma_{\text{red}} < \tau$ , the deletable channels closed) and `completeness_certified`. The latter certifies erasure only under the *worst modeled adversary* (the sup of Def. 4, i.e.  $\rho = \max_A \rho^A < \tau$ ), so a fact recoverable by the stronger reasoner is reported uncertified even when the weaker one fails; this conservative choice is why 30/81, more than any single reasoner alone flags, are uncertifiable (§5.2). A `floor_reaching` but uncertified verdict is precisely the limit result (Def. 5), now an audited field. The certificate thus surfaces both the price ( $k$ ) and the limit ( $\rho$ ) of erasure; a recall caveat (§5.6) qualifies certified verdicts, while uncertified ones are sound lower bounds.

## 5 Experiments

We evaluate on three deployed agent-memory systems spanning three architecture families: **Mem0** (Chhikara et al. 2025) (`mem0ai 2.0.7`, a deduplicating extraction pipeline over local Chroma), **Zep/Graphiti** (Rasmussen et al. 2025) (`graphiti-core 0.29.2`, a bi-temporal knowledge graph over Neo4j 5.26), and **Letta/MemGPT** (Packer et al. 2024)

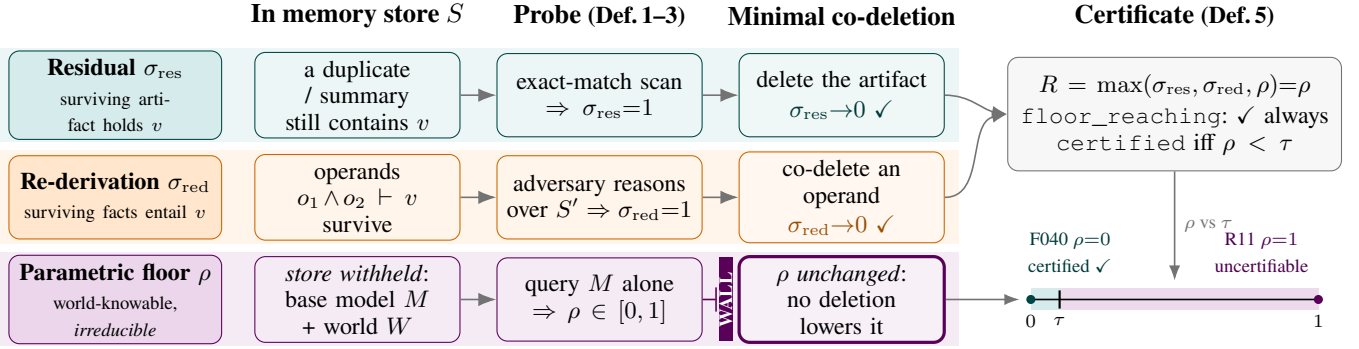


Figure 1: **The REDACT pipeline.** Recoverability decomposes into three channels (rows), each measured by a dedicated probe: *residual*  $\sigma_{\text{res}}$ , *re-derivation*  $\sigma_{\text{red}}$ , and the *parametric floor*  $\rho$ ; overall recoverability is the worst open channel,  $R = \max(\sigma_{\text{res}}, \sigma_{\text{red}}, \rho)$  (Def. 4). The greedy planner closes the two deletable channels, but  $\rho$  is a *wall* no co-deletion can lower. The certificate (Def. 5) is thus *floor\_reaching* (always achievable) yet *completeness\_certified* only when  $\rho < \tau$ . The gauge shows this limit result: F040 ( $\rho=0$ ) certifies, while R11 (“Class-3 licence  $\Rightarrow$  at least 18”,  $\rho=1$ ) does not, so  $\tau$  is an explicit policy dial.

(letta 0.16.8, an LLM-paging agent over Postgres + pgvector). Every component runs locally with pinned versions and disabled telemetry. The deletion adversary has full read access to the surviving store and black-box (no training-time) access to the base model. Wherever recovery is model-dependent it is measured against a four-reasoner adversary panel spanning prior-generation and frontier models: gpt-4o-mini, gpt-4o, Claude Sonnet 5, and GPT-5.5. The certificate takes the worst of them (Def. 4); embeddings are local all-MiniLM-L6-v2. The completeness threshold is  $\tau = 0.10$ . Datasets are synthetic subjects placed in realistic statistical contexts: 84 isolated PII facts, 92 multi-hop facts (including a multi-level set whose targets are entailed only through *unstored* intermediate facts) with their stored operands and bystanders, and 81 parametric-gradient facts across three difficulty tiers. Recovery is scored by a validated LLM judge (Zheng et al. 2023) with no observed false accepts (§5.6: 0/8 gold negatives, 95% Wilson upper bound  $\approx 0.32$ ), so under that bound *every reported recovery rate is a conservative lower bound*. Table 1 summarizes the headline numbers; results are presented in decreasing order of confidence.

### 5.1 Minimal Co-Deletion: The Planner

We run the threshold planner on all 92 multi-hop targets. For each target it probes the three channels, purges the target’s own leftover artifacts, then co-deletes entailing facts one at a time in order of entailment confidence, re-probing after each deletion and stopping the moment recoverability falls below  $\tau$ . This closes the re-derivation channel on *every* target (**100% completeness**, Wilson [0.96, 1.0]) while touching almost nothing extra: **0 spurious bystander deletions** and a **mean collateral** of  $k = 0.90$  extra facts per target (bootstrap [0.82, 0.99]). Stopping early is what keeps  $k$  low: across the run, minimality left 148 operands undeleted that a non-stopping pass would have removed, because a sufficient subset had already closed the channel.

**Why “0 spurious” is the real test.** Each target’s candi-

date pool is its true operands plus four fixed bystanders the planner must never delete, so “0 spurious” means it never once mistook a bystander for an operand (a larger fixed-bystander pool holds the same selectivity). This rests entirely on the entailment judge, and we use the stronger model deliberately. Swapping in gpt-4o-mini inflates the collateral —  $k$  rises 0.90  $\rightarrow$  1.0 and spurious deletions 0  $\rightarrow$  2 on the same targets — because, as our judge validation shows (§5.6), gpt-4o-mini wrongly calls 41.7% of *insufficient* partial operands “entailing” where gpt-4o calls none. Using gpt-4o is exactly what keeps  $k$  minimal.

**What the re-probing buys.** Our comparator, *depth-first*, is a deliberately naive upper bound: a single pass that co-deletes every candidate scoring above  $\tau$ , with no re-probing and no stopping rule. (A correctly-judged “NO” is clamped at  $0.2 > \tau$  so that a lone multi-hop operand — which cannot entail the target by itself — still stays in the pool; this makes the  $\tau$  filter effectively inert, so depth-first sweeps in nearly the whole pool.) Depth-first also reaches 100% completeness, but at  $k = 6.05$  with 342 spurious bystander deletions, because a probe-free pass cannot tell when to stop. Threshold reaches the *same* completeness at  $k = 0.90$ : the identical guarantee at  $\sim 6\times$  lower collateral (6.05  $\rightarrow$  0.90) and full selectivity (342  $\rightarrow$  0 spurious). The lower baseline is naive single-record deletion (§5.5), which removes one record and closes no re-derivation path.

### 5.2 The Parametric Floor and the Limit of Certifiable Deletion

The parametric floor  $\rho$  is the one channel deletion cannot touch: the rate at which the base model reconstructs a value from world-knowable context alone, with the store withheld entirely. We *measure* it rather than assume it, on  $n=81$  facts authored across a low/mid/high difficulty gradient. We always report the *measured*  $\rho$ , not the authored tier; the two disagree for 19/81 facts, which we keep and report by measurement, since the authored tier is only a hypothesis and the measured  $\rho$  is the evidence.

| Result               | System                  | Headline                                 | Backing control                                 |
|----------------------|-------------------------|------------------------------------------|-------------------------------------------------|
| Planner (§5.1)       | Mem0                    | 100% complete, 0 spurious, $k=0.90$      | spare-bystander + gpt-4o entailment judge       |
| $\rho$ floor (§5.2)  | base model <sup>†</sup> | 30/81 <i>uncertifiable</i> at residual 0 | worst-adversary $\rho$ ; refusal audit          |
| Re-derivation (§5.3) | Mem0, Letta             | bin1 100%, bin2 56–74%→0%                | operands-only; verified-faithful co-delete      |
| Convergence (§5.4)   | all three               | 70% / 100% (Graphiti/Letta)              | clean-delete verified first; explicit baseline  |
| Residual (§5.5)      | Mem0                    | 96.4%; 96.4%→0%                          | 2×2 embedder×cadence factorial                  |
| Judge (§5.6)         | —                       | 0/8 false-accept (CI $\leq .32$ )        | gold + 242 near-miss neg.; GPT-5.5 corroborates |

Table 1: Headline results, ordered by confidence; each “backing control” is the check that rules out the obvious confound. <sup>†</sup> $\rho$  is measured against the base model with the store withheld, so it is system-independent — the limit result; those certificates were emitted during the Mem0 run.

| field                       | COMPLETE      | INCOMPLETE     |
|-----------------------------|---------------|----------------|
| example (system)            | F040 (Letta)  | R11 (Mem0 run) |
| residual_survival           | 0.0           | 0.0            |
| rederiv_with_operands (pre) | 1.0           | —              |
| re_derivation_score (final) | 0.0           | 0.0            |
| parametric_risk_rho         | 0.0           | <b>1.0</b>     |
| facts_co_deleted ( $k$ )    | C001,C002 (2) | — (0)          |
| final_recoverability        | 0.0           | 1.0            |
| floor_reaching              | true          | <b>true</b>    |
| completeness_certified      | <b>true</b>   | <b>false</b>   |

Table 2: Two real certificates. *F040* (COMPLETE, Letta): re-derivation was open before co-deletion (rederiv\_with\_operands= 1.0) and closed after (re\_derivation\_score= 0.0). *R11* (INCOMPLETE, Mem0): residual and re-derivation are both 0, yet the base-model floor  $\rho=1.0$  forces INCOMPLETE — the limit result. Both are floor\_reaching; only F040 is completeness\_certified. Different systems, one format.

We score each fact against the *worst* of our four adversaries,  $\rho_{\max} = \max_A \rho^A$  (gpt-4o-mini, gpt-4o, Claude Sonnet 5, GPT-5.5). This splits the 81 facts into three groups (Table 3): **51 certifiable** ( $\rho_{\max} \leq \tau$ ), **12 in an intermediate band** ( $\tau < \rho_{\max} < 0.5$ ), and **18 at a hard floor** ( $\rho_{\max} \geq 0.5$ ). Averaged by authored tier, the worst-adversary  $\rho$  climbs from 0.00 (**low**) through  $\sim 0.1$  (**mid**) to  $\sim 0.6$  (**high**). The headline follows: **30 of 81 facts are not erasure-certifiable** at  $\tau=0.1$  ( $\rho_{\max} > \tau$ ) *even though their residual survival is 0 and every re-derivation path was closed*. This is the limit result: once a value is inferable from world knowledge no one can delete, no co-deletion can certify its erasure. And because the floor is a smooth gradient rather than an all-or-nothing split, the uncertifiable count moves with  $\tau$ :  $\tau$  is a deliberate policy dial, and setting it is choosing how much residual world-inferability to tolerate.

**Which adversary is worst varies by fact.** The four-model panel is not a formality. On *benign* facts the frontier models re-derive more readily, so they set the worst case there. But on *sensitive* high-tier facts — a national-service status that implies sex, a licence class that implies a minimum age — GPT-5.5 and Sonnet 5 often *refuse* (their safety filters treat the inference as disallowed: 20 refusal flags, several unanimous). A refusal makes their measured  $\rho$  a *lower* bound, so on those facts the worst adversary is instead gpt-4o. Stronger

| worst-adversary $\rho_{\max}$ | meaning           | #facts | certifiable? |
|-------------------------------|-------------------|--------|--------------|
| $\leq \tau$ ( $\tau=0.1$ )    | certifiable floor | 51     | ✓            |
| $\tau$ to 0.5                 | intermediate band | 12     | no           |
| $\geq 0.5$                    | hard floor        | 18     | no           |
| total                         |                   | 81     | 51           |

Table 3: Parametric floor  $\rho$  on  $n=81$  facts (8 samples each), binned by the worst of the four adversaries,  $\rho_{\max}$ . The floor is *graded*: a 12-fact intermediate band sits between the certifiable 51 and the hard-floor 18, so the 30/81 uncertifiable count moves with  $\tau$  (a policy dial). Measured band disagrees with authored tier for 19/81 facts.

models are therefore not uniformly stronger adversaries — exactly the adversary-relativity that taking the worst over a diverse panel is meant to capture. This positive floor does not contradict the  $\rho=0$  we report for the multi-hop facts (§5.3): there the world-knowable part is inert without the fictional stored operand and collapses under co-deletion, whereas here the context is itself world-knowable and pins the target down, so its recovery sits in the irreducible floor rather than the deletable re-derivation channel.

**Control.** By construction the  $\rho$  probe sees *only* the subject’s world-knowable context and never the stored facts, so co-deletion provably cannot lower it. Three checks harden the gradient. (i) A mid-tier numeric-tolerance sweep shows the floor persists at every tolerance — its magnitude is tolerance-dependent, its existence is not. (ii) A base-rate vs. context-lift split separates genuine inference from mere prior plausibility. (iii) A high-tier refusal audit (e.g. the sensitive item R13, refused on 4/8 samples by gpt-4o-mini) confirms that  $\rho$  on sensitive attributes is an *adversarial lower bound*. Two caveats bound the reading: because  $\rho$  is a rate over 8 stochastic samples per fact, the 30/81 count is a Monte-Carlo estimate stable to  $\pm 1$ –2 facts; and a COMPLETE certificate is complete only *modulo* the recovery judge’s  $\approx 78\%$  recall, while INCOMPLETE and aggregate figures remain safe lower bounds.

### 5.3 Decomposing Recovery: Re-Derivation at Zero Residual

To isolate re-derivation from residual survival, we use an *operands-only control*: we inject a target’s entailing facts but *never store the target value itself*, so residual survival is 0 by

construction (we verify this with the narrow deletion surface form, not the broad recovery set, so an operand that merely shares a token is not miscounted). Whatever the adversary now recovers, it recovered by *reasoning*, not by reading a leftover copy. We split the targets into two bins by what that reasoning needs: *stored-alone* (bin 1), where the surviving operands suffice on their own, and *stored+world* (bin 2), where the operands must be combined with world knowledge.

On Mem0, across all four adversaries, the target is re-derived 94–100% of the time in bin 1 and 56/59/68/74% in bin 2 ( $n=34$  per bin) — and co-deleting the operands drives both to 0% (Wilson [0, .10]), with parametric floor  $\rho = 0\%$ . Here the frontier models are the *stronger* adversaries: bin 2 re-derivation rises steadily from 56% (gpt-4o-mini) to 74% (GPT-5.5), the worst case the certificate must anticipate.

A harder *multi-level* set stresses the planner’s reach. Each target is entailed not by stored facts directly but by *derived* intermediates, which are themselves entailed by stored roots — a depth-2 join ( $A \wedge B \Rightarrow T$ ) and a depth-3 chain ( $A \Rightarrow B \Rightarrow C \Rightarrow T$ ;  $n=12$  each). All four adversaries re-derive these 100% of the time, and co-deletion again drives recovery to 0% — but only because the planner recurses *past* the target’s (unstored) direct entailers to the stored roots, which a one-hop planner would miss.

The whole control ports to Letta (exp11, same four adversaries,  $n=34$ /bin): bin 1 100%, bin 2 74–78%, again  $\rightarrow$  0% after co-deletion, with faithful direct co-deletion and bystanders both verified at 100%. One fact, F043 (a CHS $\rightarrow$ Bishan geographic leap), is a *consistent* honest negative — it leaks on neither Mem0 nor Letta, under any reasoner — which is reassuring: the probe is not simply firing on everything. Re-derivation is thus a recovery channel orthogonal to residual survival, and minimal co-deletion closes it down to the parametric floor.

**Control.** We bin recovery by mechanism and *never* aggregate across bins, and run all four reasoners because re-derivation is model-dependent (F043 above is the honest negative). On Letta, crucially, each co-deletion is issued through the *direct, deterministic* `passages.delete` (by captured passage id), not a conversational request, and we *verify* every operand passage is actually gone (faithful co-deletion = 100%, bystanders intact = 100%). So “re-derivation closed” cannot be confused with “the delete silently did not execute” — which keeps this result cleanly separate from the agent-mediated-deletion finding of §5.4.

## 5.4 Generalization: Three Families, One Phenomenon

Residual survival is not a Mem0 quirk: it appears in all three architecture families, each through a *different by-design mechanism*. We run these two cross-system probes on  $n=10$  sensitive-PII targets per system — a deliberately small, high-stakes set (see §6).

In **Graphiti**, the explicit `remove_episode` is clean at the graph level: it hard-deletes the episode, its `RELATES_TO` edges, and any orphaned entities (we verify this hard delete first). Yet the deleted value lives on in **stale entity and community summaries that the system never recomputes on deletion**: edge residue is only 30%, but

**KG/summary residue** is 70% (we count a value present in any surviving edge or summary, regardless of its bi-temporal `invalid_at/expired_at`, consistent with the full-read-access threat model). In **Letta**, deletion is agent-mediated across two surfaces — the in-context *core* blocks and the *archival* passages — and the outcome depends on how the request is phrased. An explicit dual-surface instruction is 100% **faithful**; but a *vague, deployment-realistic* request (“I’m not comfortable with you keeping X”), with the fact also sitting in archival, makes the agent scrub the core block it is reasoning over and silently miss archival: 0% **faithful**, 100% **archival residue**, 0% **core residue** (Figure 2). The agent even acknowledges a successful forget while the fact persists — an over-compliant confirmation consistent with known sycophancy (Sharma et al. 2024). Agent-mediated deletion is thus *surface-incomplete* and instruction-dependent.

**Novelty and corroboration.** This agent-loop failure is genuinely new: ForgetEval (Yang 2026) deliberately bypassed Letta’s agent loop (addressing the archival REST endpoints directly, keeping the LLM out of the recall hot path), so we test exactly the path it excluded. Conversely, ForgetEval’s own per-system notes *independently corroborate* our convergence thesis: its observation that Graphiti “sheds surface forms / synthesised fact strings” is our stale-summary residue, and that Mem0’s router prefers “link-and-keep over delete-old” is our duplication residue.

**Control.** For Graphiti, verifying the clean hard delete *first* establishes that the residue is stale summaries, not failed edge removal. For Letta, the 100%-faithful explicit baseline isolates the failure to phrasing/surface coverage, not to a missing capability.

## 5.5 Residual Survival and the Duplication Mechanism

Naive single-record deletion on Mem0 is residually incomplete for one reason: the store *silently duplicates* facts, so deleting the single surfaced record leaves copies behind. That single-record delete leaves the value present in 96.4% of isolated PII targets (81/84, Wilson [90.0, 98.8]; exp01), whereas an *artifact-aware* delete — one that purges every record carrying the value — drives residual survival from 96.4% to 0% (naive [90.0, 98.8], aware [0, 4.4]; exp02). Both arms run on the same  $n=84$  isolated PII set, whose probe values are all *globally distinct* (a cross-fact uniqueness gate enforces this), so one fact’s measurement cannot contaminate another’s. That exp01 and exp02 land independently on the same 96.4% naive rate — two separate draws of the same stochastic dedup, agreeing to within a fact or two — is itself a small replication.

**Control.** A  $2 \times 2$  factorial over embedder (MiniLM/OpenAI)  $\times$  injection cadence (0 s / 1.5 s) shows value duplication in *all four* cells, with paraphrase-variant duplicates outnumbering byte-identical ones in every cell; over the facts whose value is unique in the corpus (so vocabulary overlap cannot be miscounted as duplication), the incidence is 76–78%. The duplication is therefore a semantic-dedup *design limitation*, not an artifact of our embedder or an injection-timing race; it is independently documented in Mem0 issues #4896 (hash-only dedup),

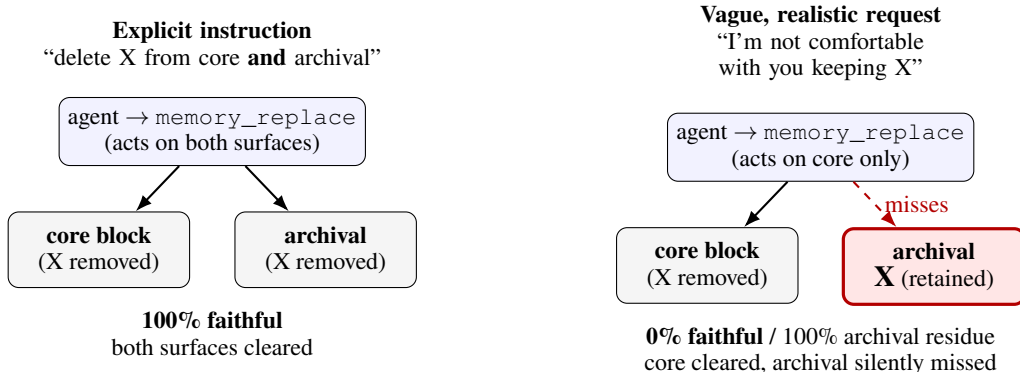


Figure 2: **Same Letta agent, opposite outcomes by phrasing.** An explicit dual-surface instruction (left) makes the agent scrub *both* the core block and the archival store: 100% faithful. A vague, deployment-realistic request (right) makes the *same* agent call `memory_replace` on the core block it reasons over and *silently miss* archival: 0% faithful, 100% archival residue. The gap is phrasing and surface coverage, not capability: the explicit baseline proves the agent *can* clear both. This is the agent loop ForgetEval (Yang 2026) bypassed by writing archival endpoints directly.

#4573 ( $\sim 37.6\%$  near-duplicates in a production audit), and #687 (the byte-identical write race). We note that these rates are *not* in tension with ForgetEval’s Mem0 numbers: their figure is forget-command success in top- $k$  retrieval, whereas ours is post-deletion recoverability by *any* channel: complementary measurements of necessary vs. sufficient conditions for completeness.

## 5.6 Judge Validation

Two LLM judges do load-bearing work — one decides whether an answer counts as *recovery*, the other whether a set of surviving facts *entails* the target — so we validate both against hand-labeled gold.

The **recovery judge** made **no false accepts** on the gold set ( $n=17$ : 7 true positives, 0 false positives, 8 true negatives, 2 false negatives;  $\kappa = 0.767$  vs. gold). Since this rests on only 8 negatives, the 0 false-accept count comes with a 95% Wilson upper bound of  $\approx 0.32$ , so the “lower bound” framing holds *up to that bound*; both of its errors are *conservative* rejections (e.g. scoring “ $\approx \text{SGD } 3,142$ ” as not recovering 3,200). To check that this pinned judge is not simply too weak to see recoveries, we re-ran the benchmark with a frontier model, gpt-5.5, as a third judge: it reproduced the gold labels *exactly* (17/17,  $\kappa=1.0$ , 0 false accepts) and disagreed with the pinned judge only on those two conservative misses — confirming the pinned judge errs only in the safe direction. Expanding the gold set with adversarial near-miss *recoveries* (“ $\approx 3,190$ ” for 3,200; “AB+” for “AB negative”) is our single most valuable validation extension (§6).

The **entailment judge** is validated on 426 pairs, including 242 near-miss negatives that should *not* entail: 12 hand-curated (topically related but with the decisive operand removed) and 230 single operands from the multi-hop corpus (a fuzzier boundary, since a lone operand can be nearly sufficient). This is where the model choice is settled. On the 12 clean curated near-misses, gpt-4o-mini false-fires on 41.7% while gpt-4o fires on *none*; across the 230 operands the false-fire rates are 62.2% (gpt-4o-mini),

39.1% (gpt-4o), and 23.9% for gpt-5.5, which we add as a third judge. gpt-5.5 is in fact the *most* accurate against the expected labels (85.2% vs. 77.7% and 64.6%) and agrees with gpt-4o at  $\kappa=0.66$  — yet on the cleanest test, the 12 curated near-misses, even it false-fires on 25% where gpt-4o fires on none. Since a false-firing entailment judge inflates the planner’s collateral, gpt-4o’s 0% curated false-fire (not model vintage) is what keeps  $k$  minimal — which is why gpt-4o is the planner’s entailment judge (§5.1).

## 6 Limitations

We state the boundaries of the evidence, leading with issues we found and corrected.

**A retracted, then powered, membership claim.** An early membership-signal result was underpowered at  $n=6$ , so we retracted it. Powered to  $n=84$  members with two matched near-twins each (168 controls), bootstrap CIs, and a label-permutation test: the intact store stays clearly detectable (AUC 0.67,  $p=.001$ , so the test has power), naive single-record deletion also leaves a detectable signal (AUC 0.67,  $p=.001$ ), and artifact-aware deletion collapses the AUC to 0.52, whose bootstrap CI [0.498, 0.552] straddles 0.5 even though a label-permutation test is marginally significant ( $p=.02$ ). Naive single-record deletion *does* leave a detectable membership signal; artifact-aware deletion attenuates it toward chance but does not provably eliminate it at this sample size.

**Judge-dependent measurements, bounded.** Both judges are validated but imperfect. On the hardest curated near-misses gpt-4o-mini false-fires on 41.7% of insufficient operands and even gpt-5.5 on 25%, so we use gpt-4o (no false-fires there) as the planner’s judge. The recovery judge had no false accepts, but over only 8 gold negatives (Wilson upper bound  $\approx 0.32$ ), so the lower-bound framing holds up to that bound; expanding the gold set with adversarial near-miss recoveries is the most valuable extension,



and COMPLETE verdicts remain modulo the recovery judge’s recall.

**$\rho$  is a lower bound where it matters most.** We estimate the parametric floor by eliciting the value from the base model, and for sensitive attributes the model often refuses (7 higher-tier items drew refusals), so the measured  $\rho$  *understates* true world-inferability for exactly the attributes a privacy audit cares about. A more willing reasoner could only push the uncertifiable count above 30/81, never below.

**The entailment graph must be known.** The decomposition and planner assume the entailment graph is specified, as it is in our controlled setup because we injected the operands. In a deployed store the graph is neither given nor easily recovered, so the planner closes only paths through *known* entailers: an unmodeled entailer is unclosable, and recovering the graph of a live memory is itself an open problem. Even at  $\rho=0$ , then, completeness is relative to the entailers the auditor can see.

**Small cross-system samples.** The convergence result probes  $n=10$  high-stakes targets per system (up from an initial  $n=3$ ). The mechanisms are qualitatively distinct and each is independently corroborated (Yang 2026), but the per-system rates (e.g. 70% KG/summary residue in the graph store) are indicative, not precise at this scale. All subjects are also synthetic, so recoveries reflect re-derivability rather than memorized real individuals.

## 7 Conclusion and Future Work

Deleting a record is not erasing a fact. We gave a causal decomposition of post-deletion recoverability, a planner that closes the deletable channels with minimal collateral, and a certificate that states, per fact, both the cost of erasure and whether it can be certified at all. Across three memory architectures the same gap recurs through different by-design mechanisms, and a *measured* parametric floor sets a hard limit on what any deletion can promise.

That limit is the paper’s broadest implication. When a fact is recoverable from world knowledge alone ( $\rho \geq \tau$ ), as 30 of our 81 gradient facts are under the strongest modeled adversary, no external-store deletion can erase it: the value is reconstructed without ever reading the store. Right-to-be-forgotten compliance then cannot be a deletion operation; it requires not storing the fact in the first place, or accepting and disclosing the residual floor. Deletion engineering closes the re-derivation and residual channels, not the parametric one.

One extension is immediate: a fourth memory architecture, to test whether the three-family convergence is universal. The deeper open problem is certifying erasure against an unbounded adversary rather than a modeled bracket: the floor we measure is a lower bound, so the true limit of forgetting in agent memory may be higher still.

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